Kane Pavlovich

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Professional Profile

PhD Researcher in Computational Neuroscience with 4+ years experience in data science. Strong expertise in; applying advanced machine learning and statistical modeling to large-scale, high-dimensional data, predictive modelling, streamlining data pipelines, and delivering actionable insights across interdisciplinary teams. Strong communicator with experience influencing research directions, mentoring teams, and presenting findings to technical and non-technical audiences.

Skills

- **Programming:** Python (Pandas, NumPy, Scikit-learn), R (tidyverse, lavaan), MATLAB, SQL, Linux/Bash, Git
- **Machine Learning:** Predictive Modelling (Regression, Classification), Feature Engineering, Ensemble Methods, Dimensionality Reduction (PCA), Time-Series Analysis, Cluster Analysis
- **Statistical Analysis:** Hypothesis Testing (A/B testing), Statistical Inference, Psychometrics, Structural Equation Modelling (SEM), Signal Processing (Fourier & Wavelet Analysis)
- **Tools & Platforms:** Tableau, High Performance Computing (Slurm), Docker, GitHub
- **Leadership:** Cross-Functional Project Leadership, Mentorship, Curriculum Development, Public Speaking

Experience

PHD RESEARCHER | MONASH UNIVERSITY | JANUARY 2022 - JULY 2025

- **Engineered and validated predictive models** (Kernel Ridge Regression, SVMs) to quantify brain-behavior relationships, **improving prediction accuracy by 15%** over previous benchmarks through novel feature engineering.
- **Developed scalable data processing pipelines** in Python and R that automated the cleaning and feature extraction of large scale (10,000+ participants) multi-modal data, **reducing data preparation time by 40%** and enabling scalable analyses.
- Applied statistical and computational methods to identify patterns and extract insights from highly complex neural and behavioural datasets, with models now adopted by international labs (Github link).
- **Presented data-driven findings** to interdisciplinary technical audiences of 200+ scientists and clinicians at 3 international conferences, and in 3 first-author papers**,** directly influencing subsequent study designs and methodologies.

HEAD TUTOR / ASSISTANT LECTURER | MONASH UNIVERSITY | FEBRUARY 2025 – PRESENT

- **Designed and delivered statistics and psychology curriculum** for 500+ undergraduates, utilizing data visualisation to translate complex concepts to non-technical audiences, **resulting in an 85% personal lecturing satisfaction rate**.
- **Led a team of 12 tutors**, implementing standardized grading rubrics and feedback systems.

TEACHING FELLOW | MONASH UNIVERSITY | FEBRUARY 2024 – FEBRUARY 2025

- Collaborated with faculty to **redesign course materials and assessments**, leading to a **15% increase** in course satisfaction scores.

Education

PHD | MONASH UNIVERSITY | SUBMITTED JULY 2025

- Titled: "*High Precision Phenotypic Estimates for Brain-Behaviour Association Analyses*" – Passed with no corrections.

MASTERS OF PSYCHOLOGY | UNIVERSITY OF AUCKLAND | AWARDED OCTOBER 2019

- Applied signal processing techniques (Fourier transforms, wavelet analysis) to model neural oscillatory dynamics *(*awarded with First Class Honours*).*

BACHELOR OF ARTS | VICTORIA UNIVERSITY WELLINGTON | AWARDED NOVEMBER 2016

Commercial Data Science Portfolio

Predicting Churn from Ecommerce Data

- **Objective:** Develop a model to identify customers at high risk of churn to enable proactive retention strategies.
- **Action & Impact:** Engineered features from user activity and complaint data. Built a multi-class decision tree classifier to segment at-risk customers. Analysis revealed that optimizing complaint follow-ups and delivery routes could reduce churn by over 50%.
- **Tools:** Python (Pandas, Scikit-learn), Exploratory Data Analysis (EDA), Decision-Trees.

Predicting Apartment Prices in the Czech Republic

- **Objective:** Create a reliable price estimation tool for first-home buyers by leveraging geographic and property data.
- **Action & Impact:** Performed geographic feature engineering and built an ensemble prediction model. Achieved 86% accuracy in predicting apartment prices based on location, square meterage, and building state.
- **Tools:** Python, Feature Engineering, Gradient Boosting, Ensemble Methods.

Clinical Risk Stratification for Cirrhosis outcomes

- **Objective:** Build a tool to help healthcare providers identify patients at highest risk of complications.
- **Action & Impact:** Processed a messy, real-world clinical dataset through robust data preprocessing and missing value imputation. Developed and compared multiple models (LightGBM, Gradient Boosting, Random Forest) to predict patient outcomes with 85% accuracy.
- **Tools:** Python**,** Data Preprocessing, LightGBM, Gradient Boosting, Random Forest.